



# Matrics Class 0+ Addendum Manual

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Matrics, Inc.  
7361 Calhoun Place, Suite 250  
Rockville, MD 20855  
Tel: 301.610.6100  
Fax: 301.610.6101  
<http://www.matrics.com>

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## Section 1. Introduction

This document serves as an addendum to the current EPC Global Standard “Class 0” Specification with respect to additional features and functions now offered in products labeled “Class 0+”. This document outlines extensions to the Class 0 specification that adds write functionality as well as some other minor functions. These extensions are implemented in accordance with the standard, and Class 0+ tags are able to be read by all Class 0 compliant readers. All information related to the EPC Global Standard for Class 0 operations can be found at the following EPC Global web site link [http://www.epcglobalinc.com/standards\\_technology/Secure/v1.0/UHF-class0.pdf](http://www.epcglobalinc.com/standards_technology/Secure/v1.0/UHF-class0.pdf).

## Acronyms and Abbreviations

The following acronyms and abbreviations are used in this document:

| ACRONYM | DEFINITION                        |
|---------|-----------------------------------|
| API     | Application Programming Interface |
| CRC     | Cyclic Redundancy Check           |
| CW      | Continuous Wave                   |
| EPC     | Electronic Product Code           |
| MSB     | Most Significant Bit              |
| RFID    | Radio Frequency Identification    |
| TBD     | To Be Determined                  |

## References

For additional information, refer to the following documentation:

- *Draft Protocol Specification for a 900 MHz Class 0 Radio Frequency Identification Tag* (available via the following link to the EPC Global web site [http://www.epcglobalinc.com/standards\\_technology/Secure/v1.0/UHF-class0.pdf](http://www.epcglobalinc.com/standards_technology/Secure/v1.0/UHF-class0.pdf))

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## Section 2. Overview

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### Document Structure

This document includes several major sections discussing topics surrounding read/write tag operations. These sections can be generally classified as follows:

- Theory and organization of operations
- Air Interface Protocol reference
- Common application examples and methods

The Theory of Operations section describes the functional use of the command level ability of the read/write tags. Details of implementation are described first in the Air Interface for the RFID reader to tag interface, and then from the application access point of view in the Reader API reference section. Finally, typical examples and their expected usage of the features and functions available under Class 0+ are described in the applications section.

## Section 3. Theory of Operations

### Memory Access

Class 0+ memory is accessed through page indexes. Class 0 defines page indexes of ID0, ID1, and ID2 but leaves undefined pages ID3..ID15. Class 0+ uses this expanded memory structure to store and retrieve data on the tag. The access is provided by commands available after singulation of the tag. Access to a page that is not defined in any particular tag will result in the loss of backscatter after an unreachable page has been received. This method is available as direct feedback from the tag as to the availability of certain sections of memory. Additionally, the 'Manufacturer ID' command response indicates to the reader which manufacturer and version number a particular tag is. The version information can be cross-referenced to provide exactly which memory pages are available and the size of each page of memory.

Matrics initial read/write products support pages ID2 (EPC code) and ID3 (user data) for the full read/write feature. Furthermore, page ID1 is supported as the last 20 bits of page ID2. ID0 is supported as is specified in Class 0 specification. Future versions of product may support additional pages, but will do so in the same manner as in page ID3. Please check with Matrics as to any additional pages on products available.

ID1 is implemented via the last 20 bits stored on the ID2 page. This information has no correlation toward the product to which the tag is attached. A randomly generated number may be stored in this location. Some of the contents of the EPC code (96 bit ID only) may overlap into these last 20 bits. Please note that if the last 20 bits of ID2 are NOT programmed, the Class 0+ tag will NOT respond per Class 0 specification. This will not affect response in ID0, ID2, or ID3 for those applications that do not utilize ID1.

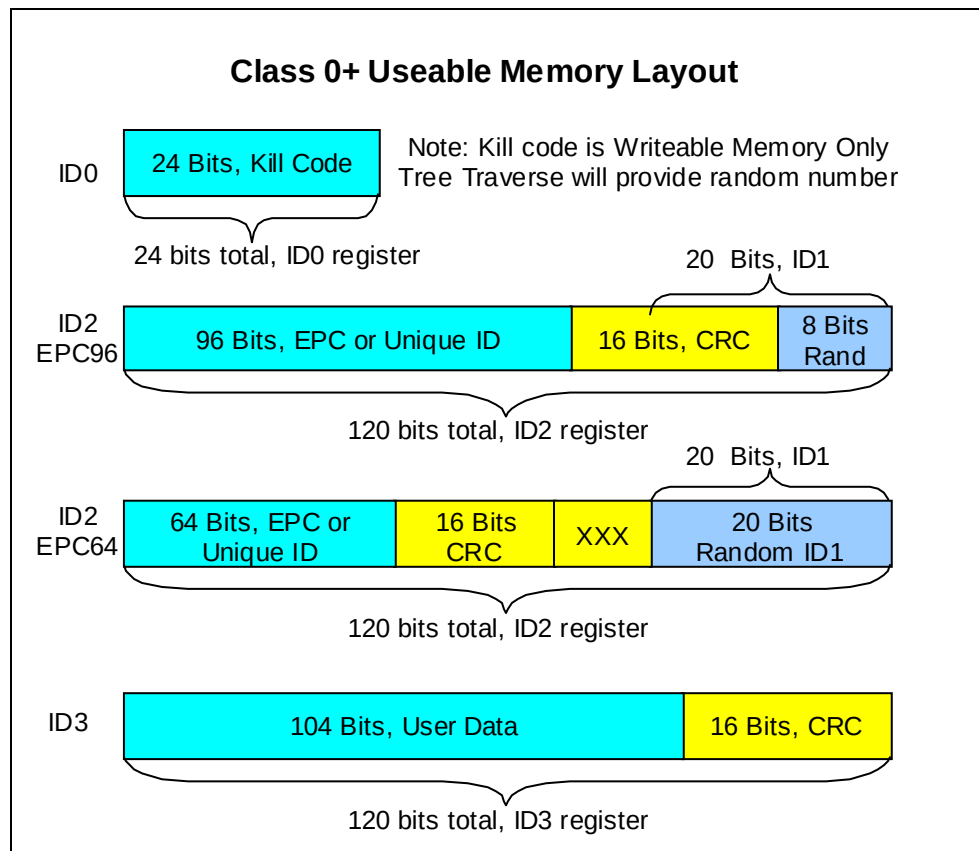
The following table and chart can be used as a cross reference for memory page index use under traversal, read, and write operations.

**Figure 1 – Available memory and usage for Class 0+ initial products**

| AVAILABLE MEMORY AND USAGE FOR CLASS 0+ INITIAL PRODUCTS |             |                     |                    |                         |
|----------------------------------------------------------|-------------|---------------------|--------------------|-------------------------|
| Page address                                             | Size (bits) | Traversal Function  | Read Function      | Write Function          |
| 0000 (ID0)                                               | 24          | Dynamic Random bits | Not applicable     | Kill Authorization Code |
| 0001 (ID1)                                               | generated   | Static Random bits  | Not applicable     | Not Applicable          |
| 0010 (ID2)                                               | 120         | EPC or Unique Code  | EPC or Unique Code | EPC or Unique Code      |
| 0011 (ID3)                                               | 120         | Not applicable      | User Data          | User Data               |



**Figure 2 – Class 0+ Useable Memory Layout**



## Memory Preparation

Class 0+ memory is written to the tag in a two-step process. First, the memory shall be erased. Then, only after erasure is the memory prepared to accept data for long-term storage. Memory on a tag that is attempted to be written without being erased may not retain the memory for the guaranteed time. Any attempts to over-write without an erase are highly discouraged.

Class 0+ implements an erase function for the erasure of a page of memory. This procedure erases a complete page within one function. Interrupting an erase operation will provide inconsistent memory results and will NOT completely erase a subset of data nor properly prepare the memory for writing. The erase procedure on the tag will give feedback to the reader upon proper completion to enable the reader to implement the next command as quickly as is possible. The erase procedure may only be implemented after singulation (one tag at a time).

Erase timing can be coordinated from the application level. Since an erase procedure takes a similar time as does the write process, tags may be pre-erased at an earlier staging level to be written faster in a more time critical fashion. An erase procedure can be implemented at any time in the tag's life. The erase will remain in effect over the tag's memory retention period. However, it is important to note that the tag memory can report random values while erased. In many applications, an erased ID2

could interfere with standard tag reading process (slow it down) and as such, erased ID2 tags should be a controlled item. Also to note, the erase procedure may not completely scramble memory and if residual memory is undesired it is suggested to erase, program tags with a zero or otherwise useless data, and re-erase to remove any residual memory effects.

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## Write Operations

Write operations are performed in Class 0+ in a bit-by-bit method. Each bit written has feedback from the tag as to success, giving the reader the ability to speed up in higher powered environments, and also detect bit failures during writes. A tag must be singulated prior to commencing the write operation. The first step is to prepare the tag for a write of a particular page. Class 0+ command 'write page' both prepares the tag to write a specific page of memory, but also returns the lock status of that page to the reader. If the page is not locked, writing bits may commence. A bit is written through the Class 0+ command 'write bit'. Status is returned from the tag to the reader to indicate when done, or if there are any errors or insufficient power while writing any particular bit. The 'write bit' command effects only the bit indexed by a current bit pointer kept on the tag. This pointer is reset to the first available bit during the 'write page' command. The pointer may be advanced at any time to the next bit index via the implementation of the 'skip bit' command. A successful write bit is not necessary to implement a skip bit. In fact, it is anticipated that a write operation to a subset of bits on a page will be necessary to some applications. Multiple, successive 'skip bit' commands may be issued in order to skip ahead into the appropriate section of a page before a 'write bit' is issued to store data.

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## Locking of Memory

In some applications, it is necessary to guard against unauthorized modification of the data written to the tag. Class 0+ adds a 'lock all' command to disable the ability to overwrite data in a particular page of memory. The 'lock all' command disables write to ALL bits of a particular page, and has this page as an argument to the command. The 'lock all' command does NOT lock all pages of memory. Each page must be locked independently. Once a page is locked, it cannot be unlocked or re-used. Applications should only lock a page after being certain that the contents of the memory match that desired and that no errors on the erase and write functions are unresolved (retried with success).

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## Kill Features

The implementation of the Kill Command is as described in the Class 0 specification. The Kill command includes a 24-bit authentication code as a command parameter in order to validate the function. Class 0+ adds the ability to store this authentication code onto the tag at any time, but most likely implemented in conjunction with the store of the EPC code into ID2. The authentication code data is stored in the memory page 0 (ID0). During the process of storing the authentication code to the tag, the authentication code is transmitted by the reader under a high power signal. It is recommended that this occur in a secure RF environment so prevent the authentication code from being disclosed to an unauthorized party in the process. The authentication code may not be read back from the tag at any time. The issuance of the kill command with an authentication code prompts the tag to verify the entire code only. At that point a match / no match condition is sent as status back to the reader.

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The authentication code, stored in page 0, can also be locked. Locking of the authentication code prevents tampering of the code. In addition, locking of the page 0 ensures an irreversible kill feature.

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## **Forced Secure Singulation**

Class 0 offers singulation methods using ID0 (true random), ID1 (static random), and ID2 (EPC). Singulation on ID2 discloses the EPC in the reader transmit signal while ID0 and ID1 modes do not. A well-engineered system can prevent the disclosure of the EPC by always singulating on ID0 or ID1. However, it may not be obvious if this system is altered inadvertently to singulate on ID2, thus disclosing the EPC codes without obvious indicators.

Class 0+ tags from Matrics offer a traversal inhibit function on the tag. When activated, the tag will no longer respond to singulation on ID2, thus only allowing singulation only on ID0 and ID1. The encoder of the tag EPC can then determine the security scheme necessary for the product to which the tag is attached. The traversal inhibit command (either in inhibit or regular mode) shall be used in every event when the ID2 is programmed. It is suggested that this command be utilized just prior to the programming of ID2. Failure to use the command will result in unpredictable tag response to ID2 traversal.

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## Section 4. Air Interface

The air interface for the additional functionality offered by Class 0+ is compatible with and in addition to the current Class 0 specification. This document will only describe additional functionality, and will not review the singulation method or standard structures available in the Class 0 specification and products.

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### Backward Compatibility

Additional features included with Class 0+ are completely backward compatible with Class 0. No commands or methods of Class 0 are removed to facilitate Class 0+. Additional commands with additional data have been added to support Class 0+. Class 0+ tags are also identifiable with standard Class 0 readers.

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### Command Mode Description

Class 0+ commands are implemented in the standard Class 0 command mode but in some cases with additional signals past the command bits. For review, commands are broken up into the following detail, bit by bit:

- For standard commands, the tag is selected by the Matrics tree traversal method defined in this document. After data (ID) bits are negotiated the next bullet item is available. For global commands, the next bullet item is only available before a complete tree traversal.
- In a standard command, the reader issues a (or multiple) data null(s). This null places this (the matching ID bits) into a command start state. For a global command, the command start state is entered for all tags in unison.
- The reader issues a data '1'. The data '1' places the tag into the command mode. After receipt of this data '1', the tag is to remain silent (NO BACKSCATTER) for this bit period.
- The reader then issues an 8-bit command, one bit at a time, most significant bit first. In addition, the reader issues a parity bit on the 8-bit command. The tag is REQUIRED to reply with an identical (confirming) data bit for each of these 9 bits. For example, if the reader sends a data '0', the tag should backscatter a data '0'.
- At this time, the tag will determine if the command is not supported, or if there was a parity error. Either situation will result in the same actions. The tag will enter either the dormant mode (invalid global command), or the command mute mode (invalid standard command). No backscatter will be allowed in these two modes by definition of these modes.
- After the 9 bits of command and parity that is implemented by this tag, the reader will follow with an undefined number of data bits that are a parameter to the command. Backscatter on these data bits is also REQUIRED for each bit as confirmation or simply data from the tag, and are to be defined for each command; listed below.

- If at any point during the command mode, the reader detects an error, it can choose to issue a data null, which will place this tag back into a command start mode. Since the data is operated on bit by bit, this may interrupt a logical operation in the middle of a data stream.

Some Class 0+ commands will require verification of or storage of information on the tag. In addition to the above rules, Class 0+ commands that implement storage or data verification shall also need to give substantial time for the tag to perform the function. This is defined as follows:

- All status verifications will conform to timing standards defined as follows:
  - The reader will issue a short period of CW (64 microseconds) to allow the tag time to do a verification process.
  - The tag may consume up to 64 microseconds (from the last data rising edge) to perform the verification function (for each status). The tag will have maximum power from the reader, in that this will occur during the CW period from the reader.
  - The reader will issue a data '1' as an indicator to return status information.
  - The tag will then backscatter the appropriate response (defined in the command detail) for the standard period of time defined for a bit interval (as defined by the T2 time point.)
  - The reader can continue with additional signals as necessary.
- All programming operations will conform to timing standards defined as follows:
  - The reader will issue a data bit for programming. Once the reader signal recovers from the modulation (CW), the cell write is allowed time to program the information. The reader shall hold CW for a substantial time period not to exceed 300 microseconds.
  - The tag may consume up to the 300 microseconds (from the rising edge) to perform the write function (for a given bit or function in process). The tag will have maximum power from the reader, in that this will occur during the CW period from the reader.
  - The reader then sends a standard data '0' bit.
  - The tag has an opportunity to reply to the reader as data on a normal reader '0' bit. The tag may respond in one of three ways.
    - No response indicates that programming has not yet been completed. The reader shall provide more CW (another 300 microseconds).
    - A data bit '0' return from the tag will indicate unrecoverable errors in programming, or power fade below minimum programming requirements during the process.
    - A data bit '1' return from the tag will indicate a success in programming
  - In this method, each tag may dynamically take as long as it is necessary to program each bit, within the maximum tolerances of 20 periods of 300 microseconds of CW plus status check bits. Substantially powered tags may take less programming time than minimally powered tags. The reader will respond faster when responding correctly to status return information from the tag during the write process.

---

## Class 0+ Command Summary

### Global commands:

|                       |                                                    |
|-----------------------|----------------------------------------------------|
| <b>ManufacturerID</b> | {11 000011 0} (reply manufacturer, version)        |
| <b>Read</b>           | {00 000110 0} (read new data from an indexed page) |

### Singulated only commands:

|                         |                                                            |
|-------------------------|------------------------------------------------------------|
| <b>Kill</b>             | {01 111111 1} (permanently disable tag)                    |
| <b>TraversalInhibit</b> | {11 000100 1} (inhibit ID page from traversal)             |
| <b>Write Page</b>       | {11 100001 0} (write data to a first available bit)        |
| <b>Write Bit</b>        | {11 100100 0} (write data to a specific bit location)      |
| <b>Lock All</b>         | {11 100011 1} (lock data on a register)                    |
| <b>Skip Bit</b>         | {11 100010 0} (skip to next available bit on current page) |
| <b>Erase</b>            | {11 100101 1} (erase all bits in the page)                 |

## ManufacturerID (binary '11 000011 0')

This command will backscatter up to 24 bits of information. The first 12 bits will be a publicly assigned manufacturer ID number. The next 12 bits will correspond to a proprietary version number.

- 6 bits, Manufacturer char 1 (see table below)
- 6 bits, Manufacturer char 2 (see table below)
- 6 bits, Product class (major version) (0-63)
- 6 bits, Product class (minor version) (0-63)

The entire code is cast into the tag design silicon. It is not stored in a read/write memory. The manufacturer and version code shall be transmitted to the reader MSB first in a process similar to the tree traversal process. As an example, for manufacturer 1, version 1, the reader shall send the command parity bit 0, the tag shall respond to this in kind with a zero. The reader shall then send a data bit 0 to start the read process. The tag will transmit the MSB of the code, in this case 0. The reader shall acknowledge with a 0. The tag will respond with its next bit if the previous bit sent matches the reader bit just received. A non-matching bit will cause the state machine to transition to either the dormant mode (in the case of a global command), or the singulated command mute state (in the case of a singulated command). The bit exchange will continue until the last bit is backscattered out (twice), or until a data null is received from the reader.

For example, the first version of the Matrics read/write tag will be M.X.2.1, from the table below, bit stream (left is first bit out): **010110 100001 000010 000001**

**Figure 3 – Manufacturer Code Characters**

| MANUFACTURER CODE CHARACTERS (FIRST 2 CHARS, 12 BITS TOTAL) |        |   |        |   |        |
|-------------------------------------------------------------|--------|---|--------|---|--------|
| 0                                                           | 000000 | G | 010000 | W | 100000 |
| 1                                                           | 000001 | H | 010001 | X | 100001 |
| 2                                                           | 000010 | I | 010010 | Y | 100010 |
| 3                                                           | 000011 | J | 010011 | Z | 100011 |
| 4                                                           | 000100 | K | 010100 |   |        |
| 5                                                           | 000101 | L | 010101 |   |        |
| 6                                                           | 000110 | M | 010110 |   |        |
| 7                                                           | 000111 | N | 010111 |   |        |
| 8                                                           | 001000 | O | 011000 |   |        |
| 9                                                           | 001001 | P | 011001 |   |        |
| A                                                           | 001010 | Q | 011010 |   |        |
| B                                                           | 001011 | R | 011011 |   |        |
| C                                                           | 001100 | S | 011100 |   |        |
| D                                                           | 001101 | T | 011101 |   |        |
| E                                                           | 001110 | U | 011110 |   |        |
| F                                                           | 001111 | V | 011111 |   |        |

**Figure 4 – ‘Manufacturer ID’ Command - Reader / Tag Interaction**

| ‘MANUFACTURER ID’ COMMAND - READER / TAG INTERACTION |          |         |                                                                                                                                                                                                                              |
|------------------------------------------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                             | Tag data | Time    | Comment                                                                                                                                                                                                                      |
| Null                                                 | None     | Std bit | Data Null starts every command                                                                                                                                                                                               |
| 1                                                    | None     | Std bit | Bit to indicate command mode                                                                                                                                                                                                 |
| 1                                                    | 1        | Std bit | First bit of command                                                                                                                                                                                                         |
| 1                                                    | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                    | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                    | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                    | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                    | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                    | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                                    | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                    | (none)   | Std bit | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command. |
|                                                      | 0        |         | Parity bit on command 0 = Normal operations - continue                                                                                                                                                                       |
| 0                                                    | 0        | Std bit | Tag response is msb of the manufacture code                                                                                                                                                                                  |
| 0                                                    | 1        | Std bit | Reader data must match last sent by tag; otherwise tag enters mute.                                                                                                                                                          |
| 1                                                    | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                    | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                                    | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                                    | 0        | Std bit | Reader has ‘010110’ from tag = first char (‘M’)                                                                                                                                                                              |
| 0                                                    | 1        | Std bit |                                                                                                                                                                                                                              |
| .....                                                |          |         | Operations continue through remaining scroll from tag                                                                                                                                                                        |

## Read (binary '00 000110 0')

The Read command has a 4-bit page argument and subsequently returns the content of that page. All data (4 bits of page data plus up to 120 bits of user data) will be sent via backscatter to the reader. The reader first sends 4 bits of page information that shall be decoded by the tag. Following these 4 bits, the reader sends out only dummy bit symbols (1 or 0), and the tag backscatters the page data on the CW portion of each dummy, regardless as to the value of the bit 0 or 1. A data '0' provides best power to the tag and longer backscatter from the tag. However, a mix of data '0' and '1' could potentially add misinformation so as to confuse a potential eavesdropper. A data null will be recognized at any time and shall terminate this command. In this way, the tag and reader maintain synchronization, while no tag data is re-broadcast by the reader.

**Figure 5 – 'Read' Command - Reader / Tag Interaction**

| <b>'READ' COMMAND - READER / TAG INTERACTION</b> |          |         |                                                                                                                                                                                                                              |
|--------------------------------------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                         | Tag data | Time    | Comment                                                                                                                                                                                                                      |
| Null                                             | None     | Std bit | Data Null starts every command                                                                                                                                                                                               |
| 1                                                | None     | Std bit | Bit to indicate command mode                                                                                                                                                                                                 |
| 0                                                | 0        | Std bit | First bit of command                                                                                                                                                                                                         |
| 0                                                | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                                | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                | (none)   | Std bit | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command. |
|                                                  | 0        |         | Parity bit on command 0 = Normal operations - continue                                                                                                                                                                       |
| 0                                                | 0        | Std bit | First bit of page data (4 bits total)                                                                                                                                                                                        |
| 0                                                | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                                | 1        | Std bit | Last bit, page index 0011 = ID3 (user data). ID0 & ID1 are not valid.                                                                                                                                                        |
| 0                                                | (none)   | Std bit | No backscatter indicates tag does not have memory ID3. Tag shall be in mute and will not backscatter until next command. Reader should discontinue command at this point                                                     |
|                                                  | b0       |         | Tag backscatters first bit of ID3                                                                                                                                                                                            |
| 0                                                | b1       | Std bit | Tag backscatters second bit of ID3                                                                                                                                                                                           |
| 0                                                | b2       | Std bit |                                                                                                                                                                                                                              |
| .....                                            |          |         | Operations continue through remaining scroll from tag ID3 memory. Reader can terminate read command at any point – a complete scroll is not necessary.                                                                       |



## Kill (binary '01 111111 1')

The kill function will permanently disable this tag from any backscatter whatsoever and is described in Class 0 specifications. In order to KILL a tag, the 24-bit code must be matched. When the reader sends 24 bits of data, the tag verifies a match, indicating match only after the 24 bits. The 24-bit authentication code is stored in the tag under page 0 (ID0). This memory location only allows information to be stored onto the tag. The tag will never disclose this code after it has been stored into memory. The ID0 location can remain unlocked, or can be locked by the Class 0+ command detailed in subsequent sections in this document.

**Figure 6 – 'Kill' Command - Reader / Tag Interaction**

| 'KILL' COMMAND - READER / TAG INTERACTION |          |               |                                                                                                                                                                                                                              |
|-------------------------------------------|----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                  | Tag data | Time          | Comment                                                                                                                                                                                                                      |
| Null                                      | None     | Std bit       | Data Null starts every command                                                                                                                                                                                               |
| 1                                         | None     | Std bit       | Bit to indicate command mode                                                                                                                                                                                                 |
| 0                                         | 0        | Std bit       | First bit of command                                                                                                                                                                                                         |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | (none)   | Std bit       | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command. |
|                                           | 1        |               | Parity bit on command 1 = Normal operations - continue                                                                                                                                                                       |
| 0                                         | 0        | Std bit       | First bit of authentication code (24 bits total), all backscattered                                                                                                                                                          |
| 1                                         | 1        | Std bit       |                                                                                                                                                                                                                              |
| .....                                     |          |               |                                                                                                                                                                                                                              |
| 0                                         | 0        | Std bit       |                                                                                                                                                                                                                              |
| 1                                         | 1        | Std bit       | Last bit of authentication code (24 bits), backscattered                                                                                                                                                                     |
| 1                                         | (none)   | Std bit       | No backscatter indicates error on tag. Reader should discontinue command at this point                                                                                                                                       |
|                                           | 0        |               | 0 indicates no auth code match. Tag will enter mute state, reader should discontinue command.                                                                                                                                |
|                                           | 1        |               | 1 indicates successful auth code match. Reader needs to supply power in order to memorialize kill function.                                                                                                                  |
| CW                                        | none     | 300 $\mu$ sec | Reader sends CW to ensure max power to tag for memory write                                                                                                                                                                  |
| 0                                         | None     | Std bit       | No response from tag indicates tag is still processing the memory write. Reader shall continue sending CW/0 combination for maximum 1000 iterations or unless data 0,1 is received.                                          |
|                                           | 0        |               | This status bit indicates insufficient power / power fade, or other errors on the tag. The reader should terminate this command and retry if this status is persistent.                                                      |
|                                           | 1        |               | A return of status '1' is an indicator of success on the kill implementation.                                                                                                                                                |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ..... | <p>Operations continue repeating the previous two items (CW / data 0) for a maximum of 1000 iterations or until a data '1' or multiple data '0's are returned by the tag. Reader may continue to issue CW/0 combinations in any case of data return from the tag. If the tag is complete, it will cease to process storage of information but will continue to backscatter a finished '1' status.</p> <p><b>The tag requires reset before the kill feature takes effect.</b> Once killed, the tag will no longer respond with any backscatter whatsoever.</p> |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Erase (binary '11 100101 1')

The Class 0+ Erase command prepares a specific page of memory for write operations only. The Erase function often does not remove residual memory values, and values may become somewhat random in nature. For a full erasure of memory, it is required to erase, write a pattern, and then erase again. It is common practice to write all '0's into a page of memory in order to remove data values from the previous write operations. The Class 0+ Erase command does give feedback from the tag to the reader to indicate when the erase has been fully completed. It is NOT possible with the Class 0+ Erase command to partially erase a page. Readers shall continue providing power until the positive feedback from the tag has been acquired.

**Figure 7 – 'Erase' Command - Reader / Tag Interaction**

| 'ERASE' COMMAND - READER / TAG INTERACTION |          |              |                                                                                                                                                                                                                                         |
|--------------------------------------------|----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                   | Tag data | Time         | Comment                                                                                                                                                                                                                                 |
| Null                                       | None     | Std bit      | Data Null starts every command                                                                                                                                                                                                          |
| 1                                          | None     | Std bit      | Bit to indicate command mode                                                                                                                                                                                                            |
| 1                                          | 1        | Std bit      | First bit of command                                                                                                                                                                                                                    |
| 1                                          | 1        | Std bit      |                                                                                                                                                                                                                                         |
| 1                                          | 1        | Std bit      |                                                                                                                                                                                                                                         |
| 0                                          | 0        | Std bit      |                                                                                                                                                                                                                                         |
| 0                                          | 0        | Std bit      |                                                                                                                                                                                                                                         |
| 1                                          | 1        | Std bit      |                                                                                                                                                                                                                                         |
| 0                                          | 0        | Std bit      |                                                                                                                                                                                                                                         |
| 1                                          | 1        | Std bit      |                                                                                                                                                                                                                                         |
| 1                                          | (none)   | Std bit      | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command.            |
|                                            | 1        |              | Parity bit on command 1 = Normal operations - continue                                                                                                                                                                                  |
| 0                                          | 0        | Std bit      | First bit of page data (4 bits total)                                                                                                                                                                                                   |
| 0                                          | 0        | Std bit      |                                                                                                                                                                                                                                         |
| 1                                          | 1        | Std bit      |                                                                                                                                                                                                                                         |
| 1                                          | 1        | Std bit      | Last bit, page index 0011 = ID3 (user data). ID1 is not valid.                                                                                                                                                                          |
| CW                                         | None     | 64 $\mu$ sec | Reader sends CW to allow the tag to check for memory page lock.                                                                                                                                                                         |
| 1                                          | (none)   | Std bit      | No backscatter indicates tag does not have memory ID3 (or other page), or that there is insufficient power available. Tag shall be in mute and will not backscatter until next command. Reader should discontinue command at this point |

|       |      |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | 0    |               | A data backscatter of '0' indicates the page is locked, and an Erase is not allowed. Reader should discontinue command at this point.                                                                                                                                                                                                                                                                                                                                                                                                   |
|       | 1    |               | A data backscatter of '1' indicates that the tag is now ready to erase this page of memory. Reader will continue with implementation of the Erase command.                                                                                                                                                                                                                                                                                                                                                                              |
| CW    | none | 300 $\mu$ sec | Reader sends CW to ensure max power to tag for memory write                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0     | None | Std bit       | No response from tag indicates tag is still processing the memory erase. Reader shall continue sending CW/0 combination for maximum 1000 iterations or unless data 0,1 is received.                                                                                                                                                                                                                                                                                                                                                     |
|       | 0    |               | This status bit indicates insufficient power / power fade, or other errors on the tag. The reader should terminate this command and retry if this status is persistent.                                                                                                                                                                                                                                                                                                                                                                 |
|       | 1    |               | A return of status '1' is an indicator of success on the page erase implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ..... |      |               | Operations continue repeating the previous two items (CW / data 0) for a maximum of 1000 iterations or until a data '1' or multiple data '0's are returned by the tag. Reader may continue to issue CW/0 combinations in any case of data return from the tag. If the tag is complete, it will cease to process storage of information but will continue to backscatter a finished '1' status.<br><b>The tag requires reset before an additional memory read command, such as read or tree traversal can take effect without error.</b> |

## Write Page (binary '11 100001 0')

The write page command is defined to set a tag index to the page specified, and sets the current bit pointer into that page as the first available bit. If the lock status for this page is unlocked, the process may continue to implement the write functions without restrictions. If the lock status for this page is locked, Write Page function is discontinued, and the Write Bit command will not be allowed for any set of bits.

**Figure 8 – ‘Write Page’ Command - Reader / Tag Interaction**

| ‘WRITE PAGE’ COMMAND - READER / TAG INTERACTION                                                                                                                    |          |         |                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                                                                                                                                           | Tag data | Time    | Comment                                                                                                                                                                                                                                 |
| Null                                                                                                                                                               | None     | Std bit | Data Null starts every command                                                                                                                                                                                                          |
| 1                                                                                                                                                                  | None     | Std bit | Bit to indicate command mode                                                                                                                                                                                                            |
| 1                                                                                                                                                                  | 1        | Std bit | First bit of command                                                                                                                                                                                                                    |
| 1                                                                                                                                                                  | 1        | Std bit |                                                                                                                                                                                                                                         |
| 1                                                                                                                                                                  | 1        | Std bit |                                                                                                                                                                                                                                         |
| 0                                                                                                                                                                  | 0        | Std bit |                                                                                                                                                                                                                                         |
| 0                                                                                                                                                                  | 0        | Std bit |                                                                                                                                                                                                                                         |
| 0                                                                                                                                                                  | 0        | Std bit |                                                                                                                                                                                                                                         |
| 0                                                                                                                                                                  | 0        | Std bit |                                                                                                                                                                                                                                         |
| 1                                                                                                                                                                  | 1        | Std bit |                                                                                                                                                                                                                                         |
| 0                                                                                                                                                                  | (none)   | Std bit | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command.            |
|                                                                                                                                                                    | 0        |         | Parity bit on command 1 = Normal operations - continue                                                                                                                                                                                  |
| 0                                                                                                                                                                  | 0        | Std bit | First bit of page data (4 bits total)                                                                                                                                                                                                   |
| 0                                                                                                                                                                  | 0        | Std bit |                                                                                                                                                                                                                                         |
| 1                                                                                                                                                                  | 1        | Std bit |                                                                                                                                                                                                                                         |
| 1                                                                                                                                                                  | 1        | Std bit | Last bit, page index 0011 = ID3 (user data). ID1 is not valid.                                                                                                                                                                          |
| CW                                                                                                                                                                 | None     | 64 µsec | Reader sends CW to allow the tag to check for memory page lock.                                                                                                                                                                         |
| 1                                                                                                                                                                  | (none)   | Std bit | No backscatter indicates tag does not have memory ID3 (or other page), or that there is insufficient power available. Tag shall be in mute and will not backscatter until next command. Reader should discontinue command at this point |
|                                                                                                                                                                    | 0        |         | A data backscatter of ‘0’ indicates the page is locked, and an write operation is not allowed. Reader should discontinue command.                                                                                                       |
|                                                                                                                                                                    | 1        |         | A data backscatter of ‘1’ indicates that the tag is now ready to write to this page of memory. Reader will continue with implementation of the Write Bit command.                                                                       |
| The Write Page also resets the tag memory pointer to the first bit in the page for the Write Bit command. The memory pointer is advanced via the Skip Bit command. |          |         |                                                                                                                                                                                                                                         |

## Write Bit (binary '11 100100 0')

The write bit command stores the data bit value passed as an argument into the tag memory. The particular bit in memory is addressed in part by the page passed as an argument, and the tag's internal register of the current bit pointer within the memory page. The write bit command verifies the status of the page lock. If the lock status for this page is unlocked, the process may continue to implement the write functions without restrictions. If the lock status for this page is locked, Write Bit function is discontinued, and the command will not be allowed.

Please note that value '0001' is undefined and should result in a command error (singulated mute state for singulated commands, dormant state for global commands). The page size in bits sent will be a total of a maximum of 128 bits for ID2 and ID3 and 32 bits for ID0. This command may require more than minimum power. If there is insufficient power to perform the write operation, backscatter will NOT occur, as backscatter is intended to be a confirmation signal of correct operation. This is allowed to occur at any bit in the process as necessary. Receiving (programming) less than 128/32 bits of information is allowed and should be expected.

The reader will send each bit as ID bits – just as it is to be sent in a future tree traversal. This is a first in first out method. The tag is to program the appropriate bit (following the timing and method described below), and then backscatter a confirmation data element (if it received a zero, it will backscatter a zero, a '1' for a '1'). The reader will end this function by issuing (sending) a data null, at which time the tag returns to command start mode. Care must be taken in that there is no protection against partial programming or programming errors on a register. To ensure uniqueness of re-addressing a particular tag for programming, it is advised to obtain an alternate page of information so that the tag can be addressed to correct programming errors.

**Figure 9 – 'Write Bit' Command - Reader / Tag Interaction**

| 'WRITE BIT' COMMAND - READER / TAG INTERACTION |          |         |                                                                                                                                                                                                                              |
|------------------------------------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                       | Tag data | Time    | Comment                                                                                                                                                                                                                      |
| Null                                           | None     | Std bit | Data Null starts every command                                                                                                                                                                                               |
| 1                                              | None     | Std bit | Bit to indicate command mode                                                                                                                                                                                                 |
| 1                                              | 1        | Std bit | First bit of command                                                                                                                                                                                                         |
| 1                                              | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                              | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                              | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                              | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                              | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                              | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                              | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                              | (none)   | Std bit | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command. |
|                                                | 0        |         | Parity bit on command 1 = Normal operations – continue                                                                                                                                                                       |
| 0                                              | 0        | Std bit | First bit of page data (4 bits total)                                                                                                                                                                                        |
| 0                                              | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                              | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                              | 1        | Std bit | Last bit, page index 0011 = ID3 (user data). ID1 is not valid.                                                                                                                                                               |

|       |        |               |                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|--------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <bit> | <bit>  | Std bit       | Single bit value <bit> is sent from reader and stored into tag buffer.                                                                                                                                                                                                                                                                                                                        |
| CW    | None   | 64 $\mu$ sec  | Reader sends CW to allow the tag to check for memory page lock.                                                                                                                                                                                                                                                                                                                               |
| 1     | (none) | Std bit       | No backscatter indicates tag does not have memory ID3 (or other page), or that there is insufficient power available. Tag shall be in mute and will not backscatter until next command. Reader should discontinue command at this point                                                                                                                                                       |
|       | 0      |               | A data backscatter of '0' indicates the page is locked, and a write operation is not allowed. Reader should discontinue command.                                                                                                                                                                                                                                                              |
|       | 1      |               | A data backscatter of '1' indicates that the tag is now ready to write to this page of memory. Reader will continue with this command.                                                                                                                                                                                                                                                        |
| CW    | none   | 300 $\mu$ sec | Reader sends CW to ensure max power to tag for memory write                                                                                                                                                                                                                                                                                                                                   |
| 0     | None   | Std bit       | No response from tag indicates tag is still processing the memory erase. Reader shall continue sending CW/0 combination for maximum 200 iterations or unless data 0,1 is received.                                                                                                                                                                                                            |
|       | 0      |               | This status bit indicates insufficient power / power fade, or other errors on the tag. The reader should terminate this command and retry if this status is persistent.                                                                                                                                                                                                                       |
|       | 1      |               | A return of status '1' is an indicator of success on the write bit implementation.                                                                                                                                                                                                                                                                                                            |
| ..... |        |               | Operations continue repeating the previous two items (CW / data 0) for a maximum of 200 iterations or until a data '1' or multiple data '0's are returned by the tag. Reader may continue to issue CW/0 combinations in any case of data return from the tag. If the tag is complete, it will cease to process storage of information but will continue to backscatter a finished '1' status. |
|       |        |               | <b>The tag requires reset before an additional memory read command, such as read or tree traversal can take effect without error following the Write Bit command.</b>                                                                                                                                                                                                                         |

## Skip Bit (binary '11 100010 0')

This simply skips the bit pointer to the next bit available in the memory page, unless the end has been reached at which point it should not advance, and will abort this function and also the write bit command.

**Figure 10 – 'Skip Bit' Command - Reader / Tag Interaction**

| <b>'SKIP BIT' COMMAND - READER / TAG INTERACTION TABLE</b> |          |         |                                                                                                                                                                                                                              |
|------------------------------------------------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                                   | Tag data | Time    | Comment                                                                                                                                                                                                                      |
| Null                                                       | None     | Std bit | Data Null starts every command                                                                                                                                                                                               |
| 1                                                          | None     | Std bit | Bit to indicate command mode                                                                                                                                                                                                 |
| 1                                                          | 1        | Std bit | First bit of command                                                                                                                                                                                                         |
| 1                                                          | 1        | Std bit |                                                                                                                                                                                                                              |
| 1                                                          | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                          | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                          | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                          | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                          | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                          | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                          | (none)   | Std bit | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command. |
| 1                                                          | 0        | Std bit | Parity bit on command 1 = Normal operations – continue                                                                                                                                                                       |
|                                                            | 0        |         | A data backscatter of '0' indicates the memory pointer exceeds the memory available on the page, and a write operation is not allowed. Reader should discontinue command.                                                    |
|                                                            | 1        |         | A data backscatter of '1' indicates that the tag is now ready to write to this next bit of memory. Reader will continue with this command.                                                                                   |

## Lock All (binary '11 10011 1')

The Lock All command locks out the entire page of memory indicated in the page address argument. Once the page of memory has been locked, no successive Write Page, Write Bit, or Erase commands can act upon this page. Locking memory (including ID0) does not prohibit a Kill function.

**Figure 11 – 'Lock All' Command - Reader / Tag Interaction**

| 'LOCK ALL' COMMAND - READER / TAG INTERACTION |          |               |                                                                                                                                                                                                                                         |
|-----------------------------------------------|----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                      | Tag data | Time          | Comment                                                                                                                                                                                                                                 |
| Null                                          | None     | Std bit       | Data Null starts every command                                                                                                                                                                                                          |
| 1                                             | None     | Std bit       | Bit to indicate command mode                                                                                                                                                                                                            |
| 1                                             | 1        | Std bit       | First bit of command                                                                                                                                                                                                                    |
| 1                                             | 1        | Std bit       |                                                                                                                                                                                                                                         |
| 1                                             | 1        | Std bit       |                                                                                                                                                                                                                                         |
| 0                                             | 0        | Std bit       |                                                                                                                                                                                                                                         |
| 0                                             | 0        | Std bit       |                                                                                                                                                                                                                                         |
| 0                                             | 0        | Std bit       |                                                                                                                                                                                                                                         |
| 1                                             | 1        | Std bit       |                                                                                                                                                                                                                                         |
| 1                                             | 1        | Std bit       |                                                                                                                                                                                                                                         |
| 1                                             | (none)   | Std bit       | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command.            |
|                                               | 1        |               | Parity bit on command 1 = Normal operations – continue                                                                                                                                                                                  |
| 0                                             | 0        | Std bit       | First bit of page data (4 bits total)                                                                                                                                                                                                   |
| 0                                             | 0        | Std bit       |                                                                                                                                                                                                                                         |
| 1                                             | 1        | Std bit       |                                                                                                                                                                                                                                         |
| 1                                             | 1        | Std bit       | Last bit, page index 0011 = ID3 (user data). ID1 is not valid.                                                                                                                                                                          |
| 1                                             | 1        | Std bit       | Single bit value <bit> is set to 1 to lock this page.                                                                                                                                                                                   |
| CW                                            | None     | 64 $\mu$ sec  | Reader sends CW to allow the tag to check for memory page lock.                                                                                                                                                                         |
| 1                                             | (none)   | Std bit       | No backscatter indicates tag does not have memory ID3 (or other page), or that there is insufficient power available. Tag shall be in mute and will not backscatter until next command. Reader should discontinue command at this point |
|                                               | 0        |               | A data backscatter of '0' indicates the page is already locked, and a new lock is not necessary. Reader should discontinue command.                                                                                                     |
|                                               | 1        |               | A data backscatter of '1' indicates that the tag is now ready to lock this page of memory. Reader will continue with this command.                                                                                                      |
| CW                                            | none     | 300 $\mu$ sec | Reader sends CW to ensure max power to tag for memory lock                                                                                                                                                                              |
| 0                                             | None     | Std bit       | No response from tag indicates tag is still processing the memory lock. Reader shall continue sending CW/0 combination for maximum 20 iterations or unless data 0,1 is received.                                                        |
|                                               | 0        |               | This status bit indicates insufficient power / power fade, or other errors on the tag. The reader should terminate this command and retry if this status is persistent.                                                                 |
|                                               | 1        |               | A return of status '1' is an indicator of success on the lock implementation.                                                                                                                                                           |
| .....                                         |          |               | Operations continue repeating the previous two items (CW / data 0) for a maximum of 20 iterations or until a data '1' or multiple data '0's are returned by the tag. Reader may continue to issue CW/0 combinations                     |



in any case of data return from the tag. If the tag is complete, it will cease to process storage of information but will continue to backscatter a finished '1' status.

**The tag requires reset before a memory read command, such as read or tree traversal can take effect without error.**

## Traversal Inhibit (binary '11 000100 1')

The utilization of the Traversal Inhibit command will disallow use of a particular page of data in general. It is generally programmed set or reset (by insisting upon a procedure in the readers) when an ID is written and after the erase function. If accidentally left un-programmed with inconsistent value, it can be later set; the tag being addressed by tree traversal on ID0 or ID1. Traversal Inhibit currently is only valid on ID2, containing the EPC or unique identifier. If a page ID2 is inhibited but requested when the reader issues any data 'null' followed by a data '0', indicating a tree traversal, the tag shall immediately enter into a mute state and proceed accordingly without backscatter. Intentionally, the only method to read information stored in the ID2 register is to extract this information using a READ command after a traversal by either ID0, or ID1. This function allows the programming of a tag to control the level of security demanded by an application.

**Figure 12 – 'Traversal Inhibit' Command - Reader / Tag Interaction**

| 'TRAVERSAL INHIBIT' COMMAND - READER / TAG INTERACTION |          |         |                                                                                                                                                                                                                              |
|--------------------------------------------------------|----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rdr data                                               | Tag data | Time    | Comment                                                                                                                                                                                                                      |
| Null                                                   | None     | Std bit | Data Null starts every command                                                                                                                                                                                               |
| 1                                                      | None     | Std bit | Bit to indicate command mode                                                                                                                                                                                                 |
| 1                                                      | 1        | Std bit | First bit of command                                                                                                                                                                                                         |
| 1                                                      | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                      | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                      | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                      | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                      | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                      | 0        | Std bit |                                                                                                                                                                                                                              |
| 0                                                      | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                      | (none)   | Std bit | No response on parity bit indicates tag does not support/recognize this command. Reader should terminate command at this point, however, in any event no more tag backscatter will result if tag does not recognize command. |
|                                                        | 1        |         | Parity bit on command 1 = Normal operations – continue                                                                                                                                                                       |
| 0                                                      | 0        | Std bit | First bit of page data (4 bits total)                                                                                                                                                                                        |
| 0                                                      | 0        | Std bit |                                                                                                                                                                                                                              |
| 1                                                      | 1        | Std bit |                                                                                                                                                                                                                              |
| 0                                                      | 0        | Std bit | Last bit, page index 0010 = ID2 (unique ID). ID0, ID1 are not valid.                                                                                                                                                         |
| <bit>                                                  | <bit>    | Std bit | Single bit value <bit> is reset to 0 to enable use of ID2 singulation, and set to '1' to inhibit use of ID2 singulation.                                                                                                     |
| CW                                                     | None     | 64 µsec | Reader sends CW to allow the tag to check for memory page lock.                                                                                                                                                              |
| 1                                                      | (none)   | Std bit | No backscatter indicates that there is insufficient power available. Tag shall be in mute and will not backscatter until next command. Reader should discontinue command at this point                                       |
|                                                        | 0        |         | A data backscatter of '0' indicates the page is already locked, and a traversal inhibit is not allowed. Reader should discontinue command.                                                                                   |

|       |      |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | 1    |               | A data backscatter of '1' indicates that the tag is now ready to set the inhibit function on this page of memory. Reader will continue with this command.                                                                                                                                                                                                                                                                                                                                                                 |
| CW    | none | 300 $\mu$ sec | Reader sends CW to ensure max power to tag for inhibit function                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0     | None | Std bit       | No response from tag indicates tag is still processing the memory store. Reader shall continue sending CW/0 combination for maximum 20 iterations or unless data 0,1 is received.                                                                                                                                                                                                                                                                                                                                         |
|       | 0    |               | This status bit indicates insufficient power / power fade, or other errors on the tag. The reader should terminate this command and retry if this status is persistent.                                                                                                                                                                                                                                                                                                                                                   |
|       | 1    |               | A return of status '1' is an indicator of success on the inhibit implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ..... |      |               | Operations continue repeating the previous two items (CW / data 0) for a maximum of 20 iterations or until a data '1' or multiple data '0's are returned by the tag. Reader may continue to issue CW/0 combinations in any case of data return from the tag. If the tag is complete, it will cease to process storage of information but will continue to backscatter a finished '1' status.<br><b>The tag requires reset before a memory read command, such as read or tree traversal can take effect without error.</b> |

## Section 5. Common Application Methods

This section serves as a structure to implement common applications with the Matrics Class 0+ tag and reader products. As a review from preceding material in this document, there are several key concepts to understand and also potential misuse of Class 0+ that may impact memory retention length and Class 0/Class 0+ existing application performance.

### Review of Key Concepts

Class 0+ basically offers rudimentary access to memory on the tag. Due to the nature of RF, and applications where tags may be in motion, there are no guarantees that a reader will stay in contact with a tag during the entire length of time to perform a requested memory function. In fact, in common applications, it is entirely likely that at some point a tag will lose signal or power enough to be interrupted mid-command, and certainly mid-application. There are key points that integrators need to keep in mind when designing applications with Class 0+:

- An erase, while preparing memory for write, does NOT necessarily delete or invalidate the contents of the memory.
- Erased memory can give random results when read back by the reader. There is no tag qualification that the memory values are valid.
- Operations can be interrupted mid-command. A command not completed can result in a memory location not holding value for the specified time period.
- The Write Page and Write Bit commands do not validate proper memory erasure. These commands may seem to work, but memory contents will not be valid for the stated life of the memory.
- When programming ID2, the application must issue a traversal inhibit command, suggested to be completed first – before the ID2 is programmed. Randomly responding tags to ID2 singulation can result if the inhibit function is not utilized.
- Once a page is locked, it cannot be unlocked.
- Once a tag is killed, it cannot be re-used.

While the list may be short and seem simple, if caution is not used, the following could result:

- Erased tags may find their way into running application space. These tags may not give valid results and may cause the readers to spend a good deal of time wasted on these tags while reducing the effectiveness of reading good tags.
- Erased tags may not appear to be erased. Typically, residual values do hold in the memory for some time after erasure. Thus, it is not possible to tell when a tag is erased. Proper handling must be in place, or erase before write (the preferred method) should be implemented at the application level.

- Applications need to keep track of which command has been interrupted and where in the overall process this takes place. A partially erased page, or partially written bit must be re-addressed so as to assure rated lifetime storage of the memory.
- The possibility that a tag thought to be erased is actually not erased. Write procedures will try to re-program the page and may seem to work for a while. These tags will likely soon develop areas in the memory that will not retain the latest programming, certainly not over the rated life of the memory. These tags will appear to fail, sometimes intermittently. Care must be taken by application design to guarantee writes only apply to erased tags.

Given the above statements, the following sections, each pertaining to a specific application, will outline safe application implementations for achieving the full potential of Class 0+ products.

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## **Application: Storing an EPC or Other Unique ID into the ID2 Memory Page**

### **Discussion and Assumptions**

The beginning assumption is that tags do not have ID2 programmed and thus do not have any uniquely identifying data to identify them. Since write operations must be after singulation, singulation must be performed on ID0 (pure random ID).

However, if singulating tags based upon ID0, there is no way to assure an application is identifying a just programmed tag. Therefore, a read back from ID2 page is recommended to assure this is not a tag that has already received it's ID2.

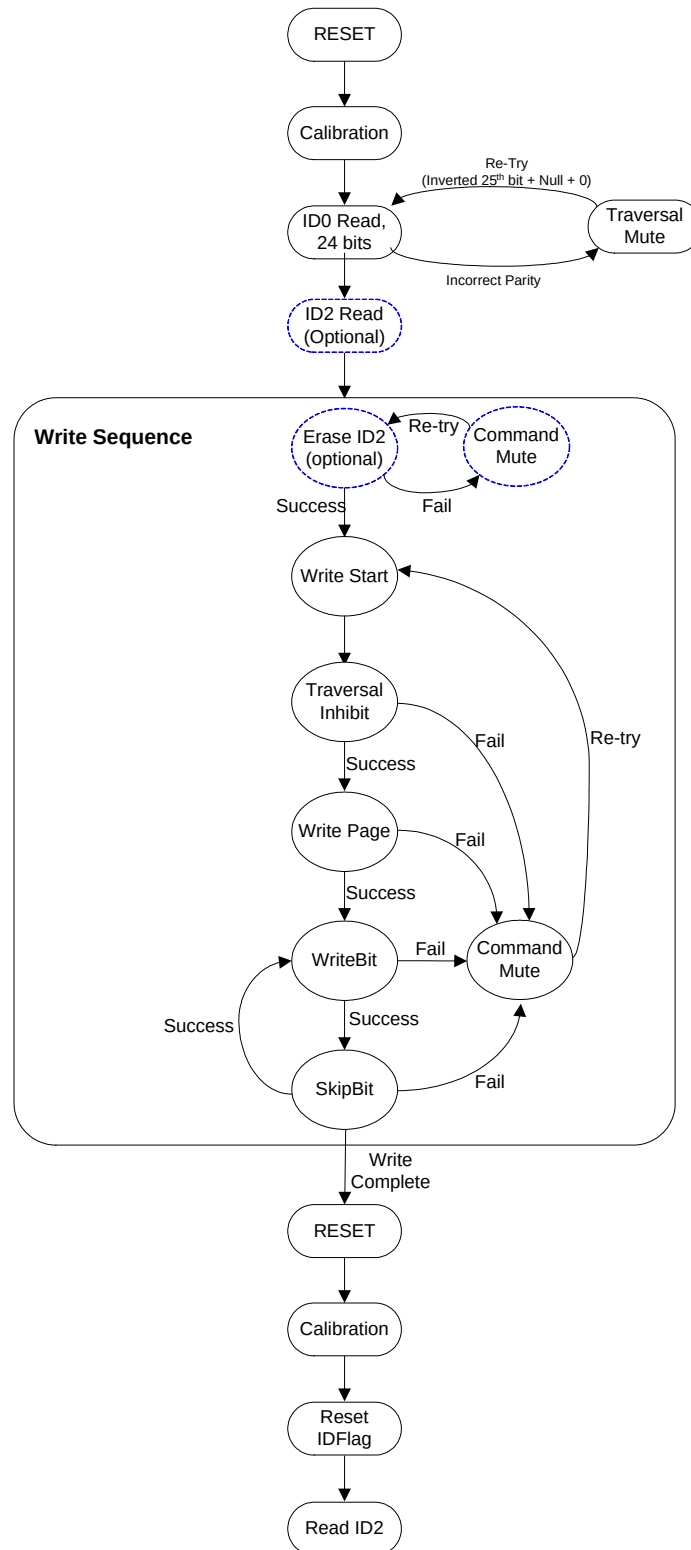
It is also recommended that an erase command be issued just prior to writing ID2 to ensure the memory is correctly prepared to write long-term information.

The write process may need to go faster, in which an erase can be done in advance. However, it is imperative that in this case no more than one tag will be present in front of a reader at a time. The potential issue is that with multiple tags, any error can cause an aborted write operation. If this happens, it is also likely that this tag cannot be re-acquired (as a result of using ID0) and hence may receive a different ID2 later on, without an erase being issued. Memory content thus will not last the rated life of the memory.

### **Procedure Flow Summary**

- Singulate first tag based on ID0 (random)
- Check ID2 to ensure not already programmed
- Erase ID2 (optional, but recommended)
- Write ID2
- Verify ID2 by re-reading memory contents.

**Figure 13 – ID2 Write Procedure Flowchart**



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## Application: Storing User Data to the ID3 Memory Page

### Discussion and Assumptions

In the case that user data is stored into memory page ID3, it is assumed that there is or can be some relationship to ID2. ID2 can be used as a logical association to ensure systematically, across readers, that a single ID3 data is stored into a particular tag. Association with ID2 also allows writes to be successful even in large tag populations. In some cases, an application may desire to correlate or even calculate an ID3 based upon a known ID2 value. In other cases, it may just serve well enough for an application to locally pair the next available ID2 with a chosen ID3 value, not having any long-term association. The association need not last any longer in an application other than to ensure that the ID3 is correctly and completely written.

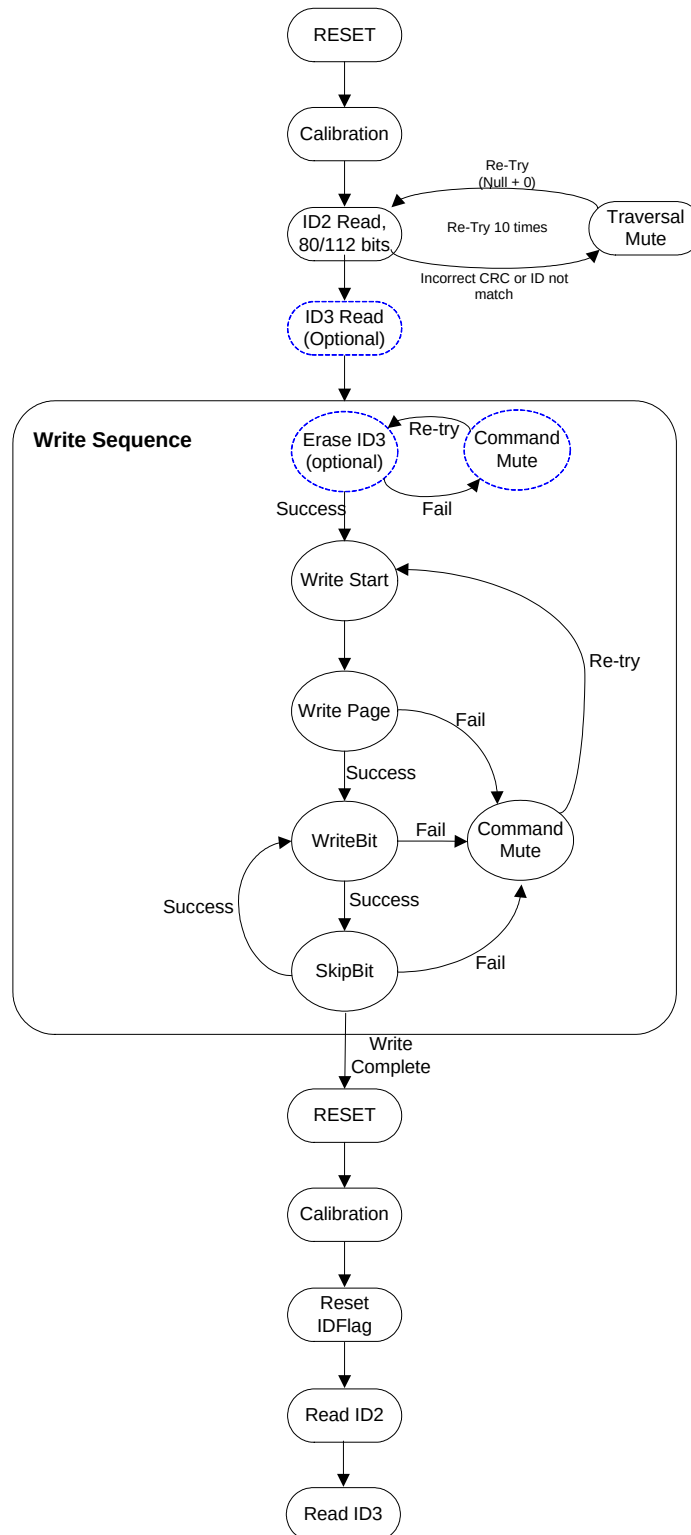
The preferred procedure is to ensure that the ID3 page is properly erased prior to writing data. Optionally, however, it may also be beneficial to first check the contents of ID3 to either: 1) ensure erasure of data does not lose that data, or 2) to perform some modification of the data as is stored on the tag vs. in a database that may not be synchronized to the actual tag data.

Partial writes into the ID3 page are allowed from the air interface, however, an application must allow for the unwritten information to be discarded even in the event that it is completely random in value. This can easily occur via hamming and/or CRC type data stored in blocks, or a first few bits defined to be the length value. An application provider may also want to consider the effects of erasure, writing of zero values, and a follow-on erasure of the ID3 register as a preparation stage separate from the actual ID3 writing process. Staging the write operations can effectively speed up the process at critical points in a moving tag environment where time becomes critical.

### Procedure Flow Summary

- Singulate first tag based on ID2
- Check ID3 to ensure it is not already programmed (optional, but recommended)
- Erase ID3 (optional, depending upon time critical applications)
- Write ID3
- Re-singulate on (same) ID2
- Verify ID3 by re-reading memory contents.

Figure 14 – ID3 Write Procedure Flowchart



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## **Application: Storing a Kill Authentication Code to the ID0 Memory Page**

### **Discussion and Assumptions**

The first assumptions that should be considered are two-fold: 1) an authentication code must be associated with an identity (ID2), and 2) an application should not allow the broadcast of the authentication code into an uncontrolled environment. The kill command, and the authentication code stored on the tag offer a great deal of protection against unauthorized killing of a tag. This is quite easily compromised if the programming of the authentication code is eavesdropped in the process of storing it onto the tag.

The Authentication code is 24 bits in length, and must be stored into the ID0 register. Reviewing air interface details shows that there is no method to read back what is stored into the ID0 register – by design so as not to compromise the authentication code. This makes the write process slightly more complicated in that a verify is not permitted. The application must ensure a clean, eavesdropping free RF environment, and shall be required to pay attention to and act upon status return bits in the writing process. Status returns indicating an error in the process should restart the process from at least the point of failure.

ID0, as the other pages, is required to be erased prior to writing. It is required (as opposed to recommended) that an erase command is implemented just prior to any write on ID0. There is no way of verifying contents of ID0, thus the application is forced to erase to be certain of long term memory of the value to be stored into ID0.

ID0, as it is limited in size compared with ID2 and ID3, will result in quicker erase command timing and status feedback – please expect this.

It is highly recommended that the page ID0 also be locked after a successful write of the ID0 / Authorization code. The lock will prevent an unauthorized modification of the code, which would render the tag open to any unauthorized use of the kill function.

### **Procedure Flow Summary**

- Singulate first tag based on ID2
- Erase ID0 (optional, depending upon time critical applications)
- Write ID0
- (Only upon success of the Write of ID0) – issue a Lock All command for ID0.



**Figure 15 – ID0 Write Procedure Flowchart**

